

HX-200 Warehouse Robot - Operator Manual

1. About this manual

This manual describes the safe operation, commissioning, maintenance and troubleshooting of the HX-200 autonomous warehouse robot. It is written for trained operators and facility technicians. Read the whole manual before using the robot for the first time, and keep it available at the site.

2. Intended use

The HX-200 is designed to move pallets, totes and racks inside warehouses and light-industrial buildings. It is not designed to carry people, to operate outdoors in rain, or to work in explosive atmospheres. Using the robot outside its intended use voids the warranty and can create safety hazards.

3. Specifications

3.1 Performance

The HX-200 has a payload capacity of 200 kg and a maximum travel speed of 2.5 m/s. It can climb slopes of up to 5 degrees and has a turning radius of 0.6 metres. The noise level during normal operation is below 62 dB(A).

3.2 Dimensions and weight

The robot is 980 mm long, 640 mm wide and 410 mm high. Its unladen weight is 118 kg.

3.3 Power

The HX-200 is powered by a 48 V lithium iron phosphate battery with a capacity of 1.5 kWh. A fully charged battery provides about 8 hours of runtime; a full charge takes 2.5 hours. The battery is rated for 2,000 charge cycles before its capacity falls to 80% of the original value.

3.4 Environment

The enclosure is rated IP54. The operating temperature range is 0 to 40 degrees Celsius, and the storage temperature range is -10 to 50 degrees Celsius. Relative humidity must stay below 90% without condensation.

3.5 Connectivity

The robot connects over dual-band Wi-Fi 6 (5 GHz recommended). An optional LTE module is available for sites without full Wi-Fi coverage.

4. Safety

4.1 Protective field

A protective field around the robot stops it within 0.5 seconds when a person enters within 1.2 metres. A warning zone of 3 metres reduces the robot's speed to 0.8 m/s when a person is detected inside it.

4.2 Emergency stop

An emergency stop button is located on the rear panel, and a second one is on the handheld controller. Pressing either button cuts power to the drive motors immediately. To release the emergency stop, twist the button and press Reset on the handheld controller.

4.3 Status lights

The light bar on top of the robot shows its state: green means ready, blue means navigating, amber means slowed by a person or obstacle, red means stopped or fault, and flashing white means charging.

4.4 Operator training

Only operators who have completed the HX Operator Level 1 course, which lasts 4 hours, may operate the robot. Technicians who perform maintenance must additionally complete the HX Operator Level 2 course.

5. Commissioning

5.1 Unpacking and first charge

Inspect the robot for transport damage before switching it on. Charge the battery for the full 2.5 hours before the first mission.

5.2 Connecting to HX-Fleet

Connect the robot to the facility network and register it in the HX-Fleet web console using the serial number on the rear label. HX-Fleet can manage up to 250 robots per site.

5.3 Mapping the facility

To create the SLAM map, drive the robot manually through the whole facility using the handheld controller at a maximum speed of 1.0 m/s. The map has a resolution of 5 cm, and a single site map can cover up to 40,000 square metres. Review the map in HX-Fleet and mark no-go zones and one-way aisles before releasing the robot for missions.

6. Navigation

The robot navigates using a 270-degree LiDAR with a range of 25 metres and a stereo camera with SLAM. Localisation accuracy is typically within 20 mm. If the robot becomes lost, it stops and asks for a relocalisation from the console.

7. Charging and battery care

The robot returns to its charging dock automatically when the battery falls below 15%. Charging is suspended if the battery temperature rises above 45 degrees Celsius and resumes when it cools down. Do not charge the robot outdoors or in direct sunlight. Battery replacement must be performed by a trained technician.

8. Maintenance

8.1 Daily checks

Before each shift, wipe the LiDAR window with a soft cloth, check that the wheels are free of debris and check that the emergency stop buttons work.

8.2 Scheduled maintenance

Perform a full inspection every 500 operating hours. The drive wheels must be replaced every 3,000 operating hours or earlier if the tread is worn below 4 mm. The robot does not need lubrication.

8.3 Firmware updates

Firmware updates are delivered over the air (OTA) and install automatically when the robot is docked. A minor release is published every quarter and a major release once per year. Robots must not be

powered off during an update.

9. Error codes

- E-12: Wi-Fi connection lost. The robot continues to the next waypoint and stops after 30 seconds without a connection.
- E-31: Stereo camera calibration lost. The robot must be recalibrated by a service technician.
- E-41: LiDAR obstruction. Clean the sensor window and restart.
- E-55: Drive motor overcurrent. Check whether a wheel is blocked and remove the obstruction.
- E-73: Battery temperature too high. Move the robot to a cool area and let the battery cool before charging. The error is raised when the battery temperature exceeds 55 degrees Celsius during operation or 45 degrees Celsius during charging.
- E-90: Emergency stop pressed. Release the button and press Reset on the handheld controller.

10. Troubleshooting

The robot does not start. Check that the emergency stop is released, that the battery is charged and that the robot is not in charging mode.

The robot moves slowly. The amber light means a person or obstacle is inside the 3-metre warning zone. Clear the zone and the robot resumes normal speed.

The robot keeps losing its position. Check for changed layouts, such as moved racks, and update the map in HX-Fleet. Clean the LiDAR window.

The robot stops in the middle of a mission with error E-12. Check Wi-Fi coverage at the location where the robot stopped and add or move an access point.

11. Transport and storage

Lift the robot only with a forklift under the marked lifting points. Store it in a dry room within the storage temperature range, and keep the battery charge between 40% and 60% if the robot will be stored for more than 30 days.